

RAGIL FIRMAN SYAH

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SUMMARY

Mechanical Engineering graduate with practical exposure to industrial maintenance and rotating equipment through an internship at PT Kilang Pertamina Internasional RU VI Balongan, complemented by hands-on experience in experimental testing and engineering projects. Experienced in centrifugal pump system analysis, piping isometric drawing interpretation, and engineering data analysis, with a strong interest in mechanical maintenance and rotating equipment

EDUCATION

Bachelor of Mechanical Engineering, Universitas Sultan Ageng Tirtayasa 08/2022 – 06/2026 | Cilegon, Banten, Indonesia
3.85/4.00

WORK EXPERIENCES

Universitas Sultan Ageng Tirtayasa 09/2025 – 07/2026 | Cilegon, Banten, Indonesia
Energy Conversion Laboratory Assistant

- Coordinated laboratory assistant activities, including task allocation, scheduling, and preparation of equipment for laboratory sessions under the supervision of laboratory management.
- Guided students through compressor performance experiments, covering test procedures, data collection, experimental result interpretation, and technical report preparation.
- Supported additional Energy Conversion Laboratory modules and gained practical exposure to the basic operation of refrigeration systems, centrifugal pumps, and diesel engines.

PT. Kilang Pertamina Internasional RU VI 04/2025 – 05/2025 | Balongan, Indramayu, Indonesia
Maintenance Execution (Rotating Equipment) Intern

- Analyzed centrifugal pump system head requirements using piping isometric drawings, compared the required head with pump capacity, and documented the findings in an internship report.
- Reviewed equipment condition using vibration reports and field observations, including blower assembly replacement following bearing failure and oil replacement on a turbine-driven centrifugal pump.
- Observed rotating equipment maintenance activities during Turnaround (TA), including centrifugal pumps, turbine drivers, compressors, blowers, and alignment procedures.

ORGANISATIONAL EXPERIENCE

Mechanical Engineering Student Association, Faculty of Engineering, Universitas Sultan Ageng Tirtayasa 06/2023 – 06/2026 | Cilegon, Banten, Indonesia

Multiple Roles — Deliberative Board, Treasury & Education

- Deliberative Board Staff Member (Mar 2025 – Jun 2026): Reviewed organizational financial reports, communicated student feedback on budget management, and contributed to the evaluation of internal financial policies and procedures.
- Treasurer (Aug 2024 – Mar 2025): Managed financial records and transaction documentation, monitored fund availability and expenditures, and prepared monthly financial reports for the executive board.
- Education Department Staff (Jun 2023 – Aug 2024): Coordinated academic support and educational programs, including exam preparation, seminars, workshops, training sessions, and student academic information.

Krakatoa Efficiency Vehicle Team (KREAMMUR), Non-Technical Staff 07/2024 – 12/2024 | Cilegon, Banten

- Prepared sponsorship proposals, followed up with prospective partners, and coordinated sponsorship and operational requirements across technical and non-technical teams.
- Supported KMHE 2024 operations through the procurement of tools, spare parts, consumables, and urgent team supplies.

SKILLS, ACHIEVEMENTS & OTHER EXPERIENCE

Hard Skills

- Rotating Equipment: Centrifugal Pumps, Rotating Equipment Maintenance Fundamentals, Equipment Alignment Fundamentals, Vibration Analysis & Balancing Fundamentals
- Engineering Analysis: Pump System Analysis, System Head & Head Loss Calculation, Piping Isometric Drawing Interpretation
- Data & Reporting: Engineering Data Analysis, Technical Reporting

Software & Tools

Microsoft Excel, AutoCAD, SolidWorks, ANSYS Fluent

Mixed-Mode Solar Dryer Performance Study, Projects (2026)

Conducted experimental testing and engineering data analysis to evaluate the effect of collector inclination on the thermal and drying performance of a mixed-mode solar dryer.

Fish Skinning Machine Prototype, Projects (2025)

Contributed to the design, fabrication, assembly, and initial testing of a semi-automatic fish skinning machine while coordinating project tasks, component selection, and documentation.